

Day 16 — Threat Intelligence APIs: VirusTotal, AbuseIPDB, Shodan

Name: Mohammed Farhan

Role: Associate – Information Security (Intern)

Organization: CyberShelter UAE

Date: 28 July 2026

Objective

Register for 4 free threat intelligence API accounts and extend Day 15's `ip_checker.py` into a full IOC enrichment tool (`ioc_enricher.py`) that queries VirusTotal, AbuseIPDB, Shodan, and AlienVault OTX for IP addresses, and VirusTotal for file hashes, producing a combined risk score suitable for real SOC triage.

Hands-On Task

API accounts registered and keys retrieved for:

- VirusTotal
- AbuseIPDB
- Shodan
- AlienVault OTX

Each key was tested individually via direct HTTP requests before any code was written, confirming genuine connectivity and authentication for all 4 services.

`ioc_enricher.py` was built incrementally, extending Day 15's script: VirusTotal integration first (tested alone), then AbuseIPDB, then Shodan, then AlienVault OTX — each tested in isolation before being added to the main enrichment loop, matching the same disciplined, step-by-step approach used throughout this internship.

Features Implemented

- IP enrichment (file-based, -f flag): reverse DNS, blocklist check, VirusTotal detection count, AbuseIPDB confidence score, Shodan open ports/ISP, AlienVault OTX pulse count
- File hash enrichment (--hash flag): VirusTotal Malicious/Suspicious/Harmless engine counts
- Combined risk score: rated HIGH if the IP is blocklisted, OR VirusTotal shows any malicious detections, OR AbuseIPDB score exceeds 50, OR OTX pulse count exceeds 5; MEDIUM if no reverse DNS record exists; otherwise LOW
- Exponential backoff: applied to VirusTotal and AbuseIPDB requests, automatically retrying with increasing wait times (1s, 2s, 4s...) on HTTP 429 rate-limit responses
- Edge case handling (carried forward and extended from Day 15): invalid IPs, DNS timeouts, missing input files, and now malformed API responses

Verification

Test 1 — IP file enrichment:

```
python ioc_enricher.py -f test_ips.txt
```

Confirmed correct output across all 4 threat intelligence sources for 3 test IPs, with accurate risk classification (blocklisted IP correctly flagged High; clean, trusted IPs correctly flagged Low).

```
PS C:\Users\FARHAN\Desktop\day16> python ioc_enricher.py -f test_ips.txt
```

IP	Reverse DNS	In Blocklist	VT Detection	AbuseIPDB Score	Shodan Info	OTX Pulses	Risk Level
8.8.8.8	dns.google	NO	0 malicious	0	2 ports (443,53), Google LLC	0	Low
1.1.1.1	one.one.one.one	NO	0 malicious	0	11 ports (161,2082,2083,2086,2087,80,8880,8080,53,8443,443), Cloudflare, Inc.	0	Low
192.168.1.1	etisalat-s3	YES	0 malicious	0	Error: Invalid IP	0	High

Test 2 — File hash enrichment:

```
python ioc_enricher.py --hash 44d88612fea8a8f36de82e1278abb02f
```

Used the well-known EICAR test file hash; confirmed VirusTotal correctly identified it as malicious (66 detections), demonstrating the hash-lookup feature functions correctly against a genuine, known reference value.

```
PS C:\Users\FARHAN\Desktop\day16> python ioc_enricher.py --hash 44d88612fea8a8f36de82e1278abb02f
File Hash Enrichment Result
```

Hash	Malicious	Suspicious	Harmless
44d88612fea8a8f36de82e1278abb02f	66	0	0

```
PS C:\Users\FARHAN\Desktop\day16> 
```

Deliverable

- ioc_enricher.py — working command-line tool, committed to Git repo (week-4/day-16-deliverables/)
- README.md — usage instructions, risk logic explanation, and documented limitations

Conclusion

Day 16 extended Day 15's foundational script into a genuinely functional IOC enrichment tool, integrating 4 real external threat intelligence services rather than simulated data. Building the script incrementally — testing each API integration in complete isolation before adding it to the main tool — allowed issues to be caught and resolved one at a time rather than debugging several new integrations simultaneously. This tool is directly reusable for real SOC triage work and forms the foundation for Day 17's continued IOC enrichment development.